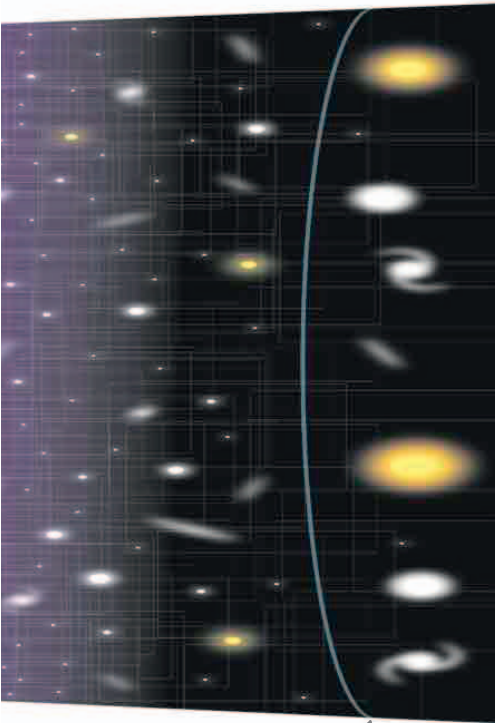
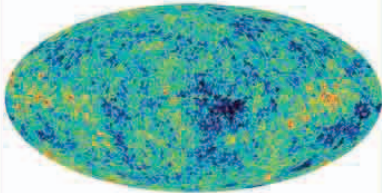


REAL WORLD

Cosmic microwave

Microwave radiation from the Big Bang was discovered by accident by radio astronomers who thought the radiation was background noise. Today, satellites have made very detailed maps of the radiation, which show slight temperature variations (shown as different colors below), revealing slight variations in the density of the young Universe.



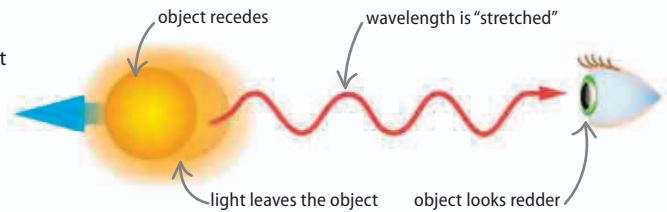
Present
The Universe as it is today, 13.7 billion years old.

Red shift

Galaxies exist in clusters, and the key piece of evidence for the Big Bang is that these clusters are all moving apart—in other words, the Universe is expanding. Astronomers know this because they can split the light from the galaxies into spectra, which are like rainbows containing lines of light or dark that show the substances present in them. In spectra from distant galaxies, the positions of the lines are all shifted to longer wavelengths—that is, toward the red end of the spectrum.

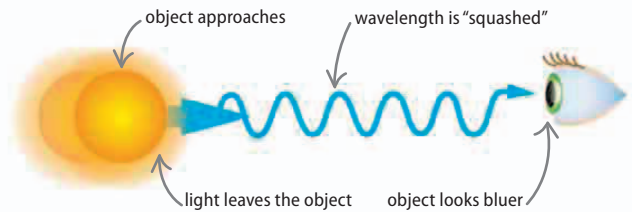
▷ Red shift

The light from a star, galaxy, or other bright object is reddened very slightly if it is moving away from the observer.



▷ Blue shift

When an object approaches the observer, its light shifts the other way, toward the blue end of the spectrum.



What is out there?

Stars and galaxies are just a tiny part of the total mass of the Universe. Most is made up of forms of matter and energy that cannot be seen. Galaxies contain invisible dark matter, which scientists know is there by observing galaxy behavior. There is also an unknown force accelerating the expansion of the Universe, known as dark energy.

▽ Universal matter

These pie charts show the make-up of the Universe. The atoms that make the stars and galaxies are mainly hydrogen and helium. The additional heavier elements are made within stars. Those beyond iron (see pages 116–117) form when massive stars explode.

